

Observation

The couple eliminated in each (season, week); seasons 3-27

each (season, week) gives one linear inequality on u

Constraint (combination rule)

score = $\alpha * \text{judge} + (1 - \alpha) * \text{fan}$; the lowest score is eliminated

share from u ; LP inverts all weeks to a feasible u

Latent variable u (one per season)

$u_{i,s}$ is non-negative and shared by all weeks; share_i
= $u_i / (\text{sum over the in-panel contestants of } u)$